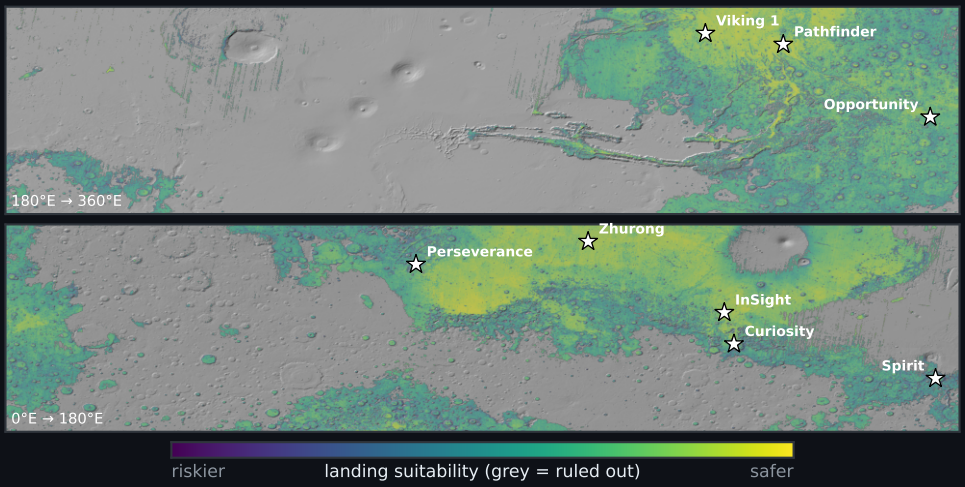
PLANETARY GIS PROJECT

Can an Earth GIS method choose where to land on Mars?

I took the same multi-criteria analysis we use on Earth to pick a site for a road, a farm or a building, and applied it to the whole equatorial belt of Mars using free NASA/ESA orbital data. Then I checked it against the 8 places where real landers actually touched down.



Result: all 8 of 8 real landing sites scored in the top 20 % of the map.

15 slides · method, results, and the part that surprised me (slide 12).

Mohamed Jihed Fradi · Geomatics Engineering student, MSE Manouba

THE PROBLEM

Why choosing a landing site is hard

IN PLAIN WORDS

A Mars lander has about 7 minutes to go from 20,000 km/h to zero, on its own, with no pilot. It can't choose where it lands precisely: it comes down somewhere inside an oval several kilometres wide. So the whole oval has to be safe.

- ▲ **Steep slopes**
  the lander can tip over, and its radar gets confused
- ▲ **Rocks and rough ground**
  a single big rock can break a leg or puncture an airbag
- ▲ **High ground**
  less air above it to slow the parachute down
- ▲ **Deep dust**
  soft ground that can't support the lander or a rover
- ▲ **Craters**
  rims, boulders and steep walls everywhere
- ▲ **Far from the equator**
  colder, and less sunlight for solar panels

NASA and other agencies spend years on this, with teams of scientists and engineers. My question: how close can a transparent, reproducible GIS model get, using only public data?

TECHNICAL DETAIL

Perseverance's landing ellipse was ~7.7 × 6.6 km (with Terrain-Relative Navigation); Viking's in 1976 was ~280 × 100 km. Entry, descent and landing (EDL) takes ~7 min. This project uses a minimum candidate-zone size of 50 km², about one modern landing ellipse.

THE IDEA

Multi-criteria analysis: stack maps, score, combine

IN PLAIN WORDS

On Earth, to find the best site for something, GIS analysts stack several maps (slope, soil, distance to roads…), give every location a score on each one, decide how much each map matters, and add the scores up. I did exactly that for Mars:

1. **Collect the data**
   3 free datasets measured from orbit
2. **Put everything on one grid**
   76 million cells of 1 km × 1 km
3. **Build 6 criteria**
   slope, roughness, height, dust, craters, latitude
4. **Score 0 → 1 and rule out the impossible**
   too high, too dusty or too steep = out
5. **Weigh the criteria (AHP)**
   structured pairwise comparisons
6. **Combine, then test against real landers**
   8 real sites vs 1,000 random spots

TECHNICAL DETAIL

Method: Analytic Hierarchy Process (Saaty, 1980) + weighted linear combination, a standard GIS multi-criteria decision analysis (MCDA). Everything is code: Python, GDAL/rasterio, NumPy/SciPy, GeoPandas; 4 Jupyter notebooks and 27 automated tests, all on GitHub.

STEP 1 · DATA

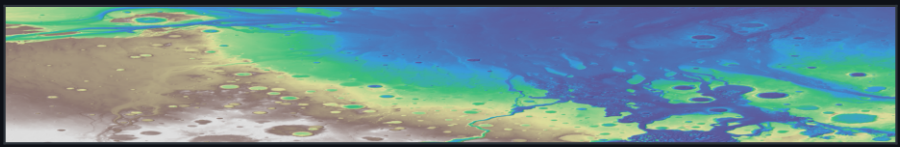
Three free datasets, all measured from orbit

IN PLAIN WORDS

Every map below covers the same piece of Mars (Chryse Planitia, where Viking 1 and Pathfinder landed). Each one tells us something different about the ground.

Height (elevation)

laser altimeter + stereo camera



blue/green = low ground · brown/white = high ground

Dust vs. solid ground

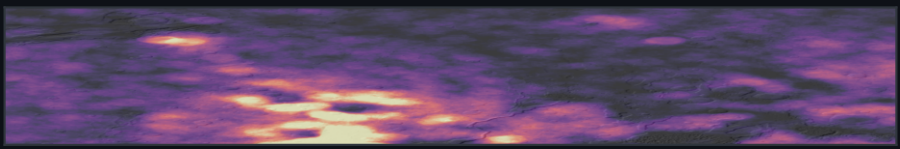
heat sensor: how fast the ground cools at night



dark blue = soft dust · yellow = solid, rocky ground

Crater density

384,343 catalogued craters ≥ 1 km



dark = few craters · bright = many craters

TECHNICAL DETAIL

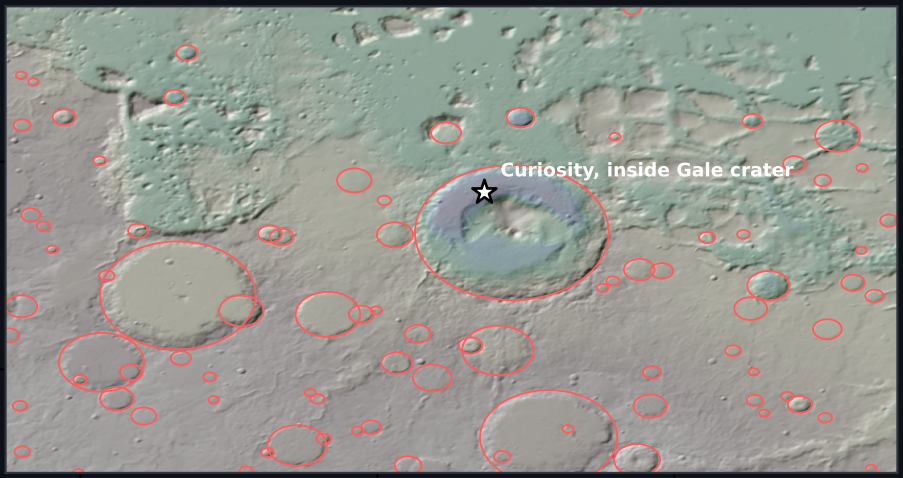
MOLA–HRSC blended DEM v2 (USGS, 200 m/px; Fergason et al. 2018) · TES nightside thermal inertia (Putzig & Mellon 2007, ~3 km/px; < 100 J m $^{-2}$ K^{-1} s$^{-½}$ ≈ deep dust) · Robbins & Hynek (2012) crater database. The 11 GB DEM is streamed over HTTP; only the ±30° band is read.

STEP 2 · ONE COMMON GRID

Making every map line up, pixel for pixel

IN PLAIN WORDS

The datasets come from different instruments, with different pixel sizes and even different longitude conventions. Before comparing them, I re-projected everything onto one grid of 1 km cells covering the band 30° north to 30° south of the equator. Check: the red circles are craters from the catalogue, drawn on top of the elevation map. They sit exactly on the crater rims, so the layers are aligned.



TECHNICAL DETAIL

CRS: Mars 2015 equirectangular, IAU_2015:49910 (sphere R = 3,396,190 m), planetocentric latitude, east-positive longitude. Grid 21,338 × 3,556 px. In this projection an east–west pixel covers only cos(latitude) km on the ground (866 m at 30°), so slope, crater kernels, zone areas and random sampling are all corrected for it (otherwise slope is off by up to 13 %).

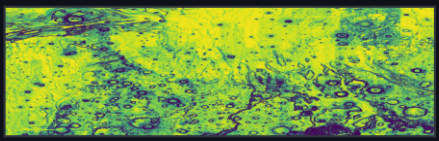
STEP 3 · SIX CRITERIA

Six questions asked of every 1 km cell

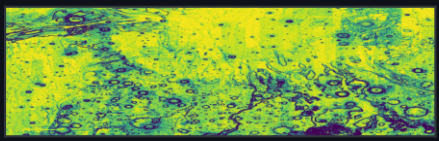
IN PLAIN WORDS

From the data I computed six maps. Each one answers a simple question. Below, each map is already converted to a score: yellow = good for landing, purple = bad.

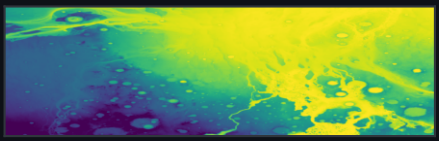
How steep is it?
flatter is better



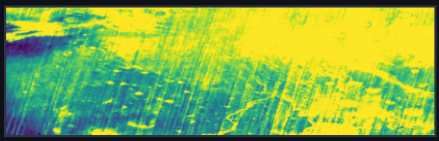
How bumpy is it?
smoother is better



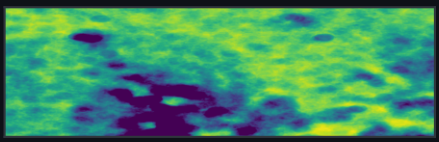
How high is it?
lower is better



Dust or solid ground?
solid is better



How many craters nearby?
fewer is better



How far from the equator?
closer is better



TECHNICAL DETAIL

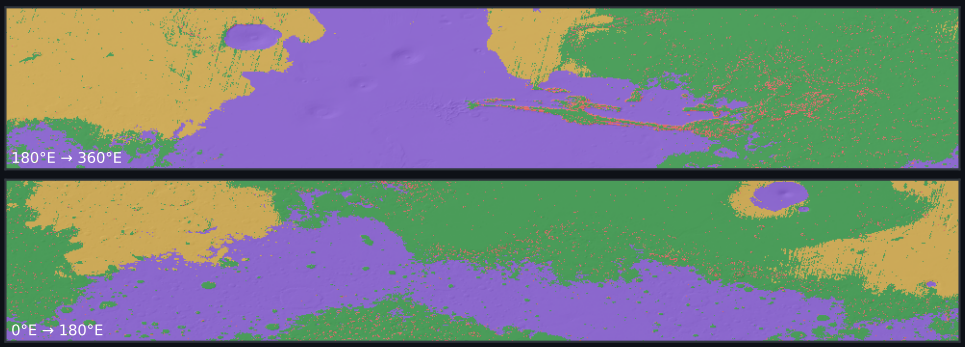
Slope from the latitude-corrected DEM gradient; roughness = Terrain Ruggedness Index (mean absolute height difference to the 8 neighbours); crater density = craters ≥ 1 km per 10,000 km² within 50 km (elliptical kernel in pixel space, catalogue read to ±31° to avoid edge effects); |latitude| from the grid. Same region as slide 4. At 1 km, roughness closely follows slope (visible above); a 200 m roughness layer is the planned improvement.

STEP 4 · SCORES AND HARD LIMITS

Grade everything 0 → 1, and rule out the impossible

IN PLAIN WORDS

Each criterion is turned into a grade from 0 (worst) to 1 (best), like marking an exam. Some ground is simply impossible, whatever its other grades, so it's removed first. Only 39% of the band survives:



- possible (39%)
- too high: above +1 km (38%)
- too dusty: soft, deep dust (20%)
- too steep: over 10° (2%)

Big volcanoes (Tharsis, Elysium) are out because they're too high; the bright, dusty plains of Arabia and Tharsis are out because the surface is too soft.

TECHNICAL DETAIL

Linear fuzzy membership: elevation 1 at ≤ −4 km → 0 at ≥ +1 km; slope 1 at 0° → 0 at ≥ 5°; thermal inertia 0 at ≤ 100 → 1 at ≥ 300; roughness and craters 1 at the minimum → 0 at the 95th percentile of possible ground (TRI 88 m, 74 craters/10^4 km²); latitude 1 at 0° → 0 at 30°. Hard constraints: elevation > +1 km, slope > 10°, TI < 100.

STEP 5 · WEIGHTS (AHP)

Which criteria matter most?

IN PLAIN WORDS

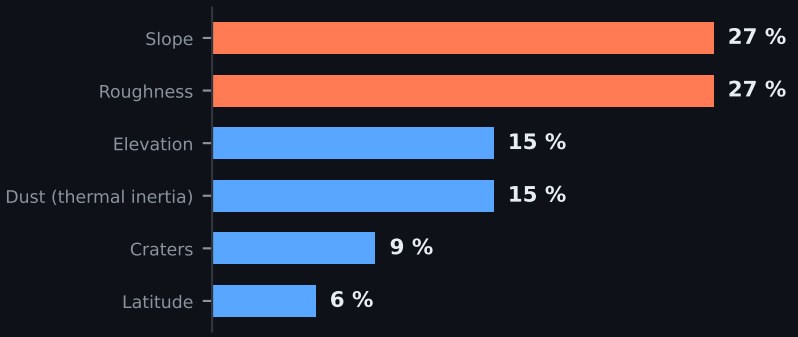
Instead of guessing percentages, I used a method called AHP: compare the criteria two at a time and say how much more important one is than the other. For example:

Slope vs Elevation — **2× more important**
a tip-over is immediate; high ground only reduces margins

Slope vs Craters — **3× more important**
craters mostly matter through the slopes they create

Slope vs Latitude — **4× more important**
within ±30°, latitude is a preference, not a danger

15 comparisons like these are turned into weights by matrix maths, and the method checks that my answers don't contradict each other.



TECHNICAL DETAIL

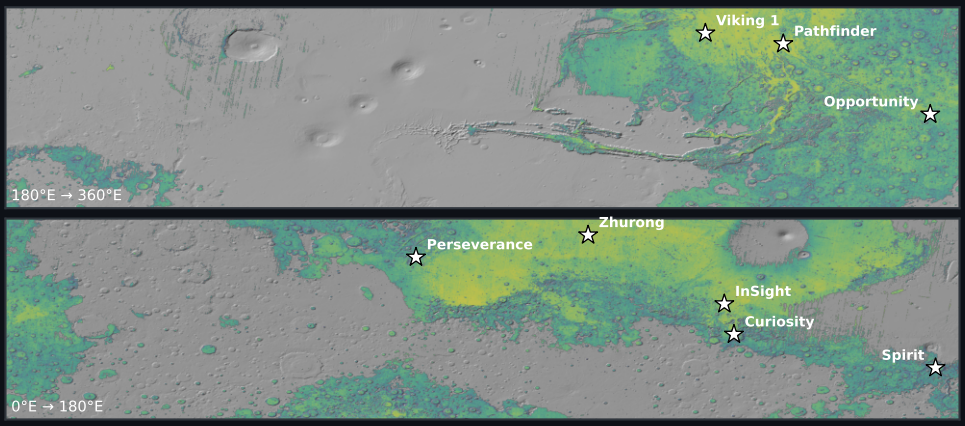
Saaty 1–9 scale, 6 × 6 reciprocal matrix; weights = principal eigenvector. λmax = 6.041, consistency index CI = 0.0083, consistency ratio CR = CI / RI = 0.0067 (RI = 1.24 for n = 6). CR < 0.10 means the judgements are consistent. Safety criteria dominate: a failed landing ends the mission.

STEP 6 · THE RESULT

The landing-suitability map of equatorial Mars

IN PLAIN WORDS

Each cell's final score is the weighted sum of its six grades: 1 would be perfect on everything. Yellow areas are the safest places to land; grey areas were ruled out. Stars are the 8 real landers.



Two huge low plains stand out: Chryse Planitia (top strip, right) and the Isidis–Elysium–Utopia lowlands (bottom strip). That's exactly where most landers went.

TECHNICAL DETAIL

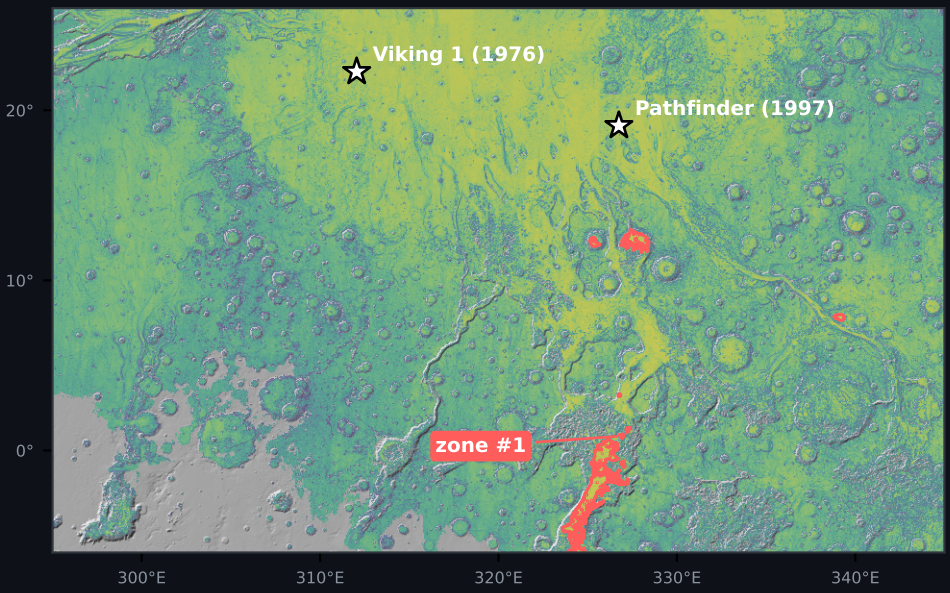
$S = \Sigma\, w_i \cdot s_i$ over the 6 criteria, set to 0 where a hard constraint fails. Output: a 1 km GeoTIFF (21,338 × 3,556). Classified into 5 quantile classes; the "very high" class ($S \geq 0.80$) feeds the candidate-zone search on the next slide.

CANDIDATE SITES

Zooming in: the best landing zones

IN PLAIN WORDS

I grouped the best-scoring cells into zones and kept only those at least as big as a modern landing oval (50 km²). The 10 best zones are outlined in red. Most sit on the flat floors of ancient flood channels (Ares and Tiu Valles), just south of where Pathfinder landed in 1997.



TECHNICAL DETAIL

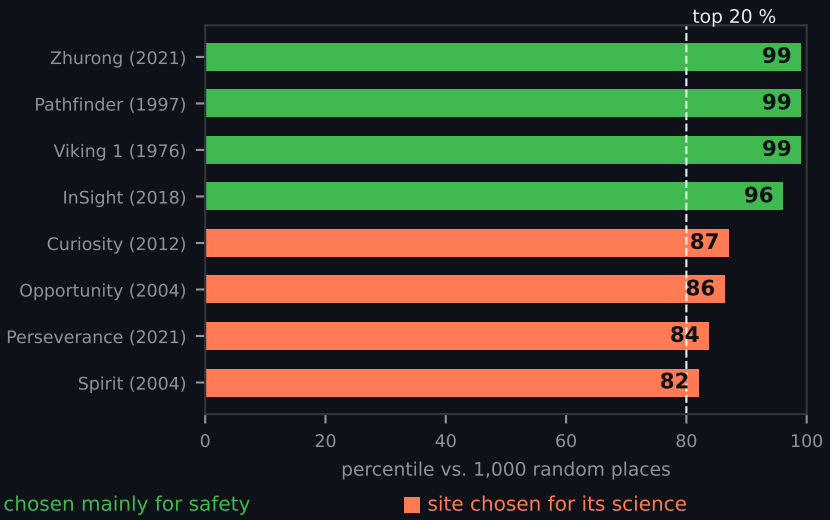
Zones = 8-connected cells with S ≥ 0.80 (merged across the 180° seam), area from true cell areas, minimum 50 km², ranked by area-weighted mean score. Zone #1: 0.85° N, 326.92° E, 138 km², mean score 0.90. Exported as a GeoPackage.

THE TEST

Does the map agree with where real landers went?

IN PLAIN WORDS

If the model makes sense, the 8 places where landers really touched down should score much higher than random spots on Mars. So I compared each landing site with 1,000 random locations. A score of 90 means "better than 90 % of random places".



All 8 landing sites are in the top 20 %. If the map were random, the chance of this would be about 1 in 100,000.

TECHNICAL DETAIL

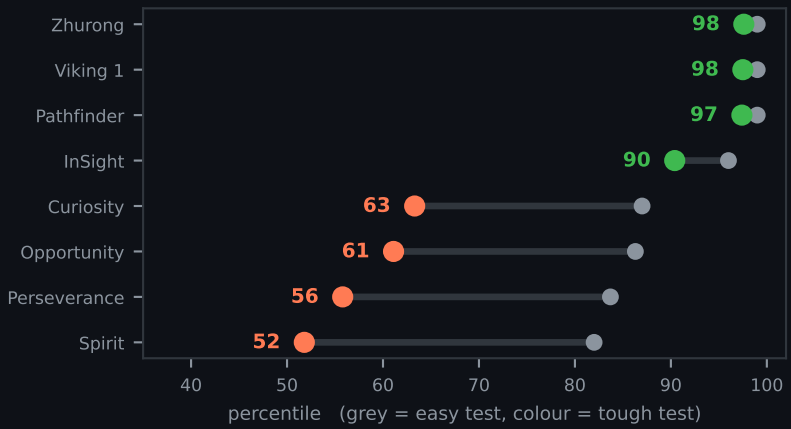
Each site = mean score within 10 km. Random points drawn uniformly by area (weighted by cos latitude) over the whole band, measured the same way. One-sided Mann–Whitney U test: p = 9.0e-06. Coordinates are planetocentric, from PDS/NASA sources.

THE SURPRISE

The "misses" show where science beat safety

IN PLAIN WORDS

A tougher test: compare each site only with ground that was already possible (no volcanoes, no dust fields). Now the 8 sites split cleanly in two. The four chosen mainly for safety (Viking 1, Pathfinder, InSight, Zhurong) stay at the very top. The four rover sites chosen for their geology (Gusev, Meridiani, Gale, Jezero) drop to the middle.



Scientists accepted less safety margin to reach interesting rocks, helped by better landing technology (airbags, then guided entry, then Terrain-Relative Navigation). The model only knows about safety, so the gap measures the margin traded for science.

TECHNICAL DETAIL

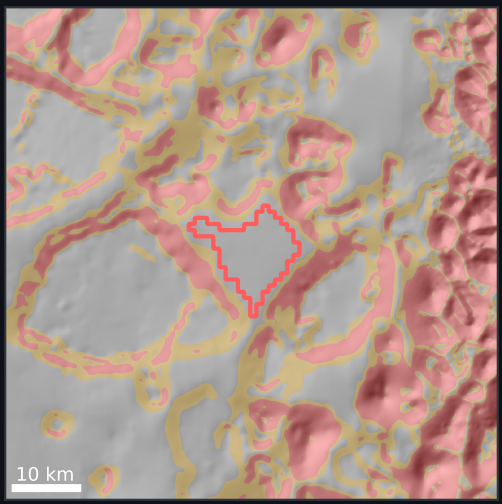
Reference = 1,000 random points drawn only from constraint-passing ground. 4 of 8 sites stay in the top 20 % (one-sided Mann–Whitney p = 0.0035). Safety-first sites: 90–98th percentile; science-first: 52–63th. Grouping = primary driver of the final site choice (simplified).

REALITY CHECK

Looking at the #1 zone 5× more closely

IN PLAIN WORDS

The model works with 1 km cells, which can hide cliffs. So I re-downloaded the #1 zone at 200 m resolution and measured its slopes. Inside the red outline, the ground is almost perfectly flat (typical slope 0.7°, no spot above 5°), while the channel walls around it are steep. The zone passes.



■ slope 5–10° ■ slope > 10° ▢ zone #1 outline

TECHNICAL DETAIL

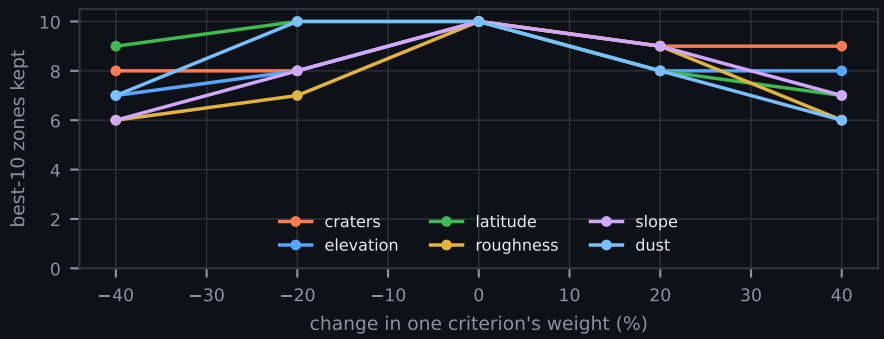
200 m DEM window (80 × 80 km) cut from the remote MOLA–HRSC file. Inside zone #1: slope median 0.72°, 95th percentile 1.89°, 0 % of cells > 5°. Whole window: median 5.8°, 95th percentile 22.5° (channel walls). Location: 0.85° N, 326.92° E, Ares Vallis floor.

ROBUSTNESS

What if my weights were wrong?

IN PLAIN WORDS

The weights from slide 8 are my judgement, so I tested how much the answer depends on them. I changed each weight up and down by 20 % and 40 % and re-ran everything, and then ran 500 versions with random weights. The best zones mostly stay the same, and the real landing sites score high every time.



500 random weightings: how often is each place in the top 10 %?



TECHNICAL DETAIL

One-at-a-time: ±20 % keeps 7–10 of the 10 best zones; ±40 % keeps 6–10. Scenarios: "Safety first" (slope + roughness 70 %) and "Equal weights" keep only 4, 3 of the 10 zones, but the mean landing-site percentile stays at 90 and 93 (baseline 92). Monte Carlo: 500 Dirichlet draws centred on the AHP weights. Ranking is sensitive; the map is stable.

TAKEAWAYS

What I learned, and what's next

Earth GIS methods transfer to other planets.

Projection, resampling, raster algebra and MCDA work on Mars exactly as on Earth, once you respect its own coordinate system.

Validation is where the insight is.

The simple test said "8/8, great". The tougher test revealed the safety-vs-science trade-off, which is the real story.

Honest limits.

1 km cells can't see individual rocks; the thresholds are illustrative, not mission specifications; real site selection also uses rock counts from HiRISE images, radar and wind data.

Next.

200 m roughness for the whole band, a science-value layer from the geological map, and the same analysis for the Moon's south pole (Artemis).

Code, data sources, notebooks and all figures:

github.com/MJFradi/mars-landing-site-suitability

Questions and feedback welcome, especially from planetary scientists and GIS people!

TECHNICAL DETAIL

Data: NASA/ESA/DLR/USGS (MOLA–HRSC DEM v2), Putzig & Mellon 2007 (TES thermal inertia), Robbins & Hynek 2012 (craters). Tools: Python, GDAL/rasterio, NumPy, SciPy, pandas, GeoPandas, Matplotlib, Jupyter. MIT licence. Fully reproducible: one download script, 4 notebooks, 27 tests.